(close to 1) probabilities without underflowing or overflowing the computer's numerical precision. Logarithms also have nicer mathematical properties that we won't cover, but that is why you'll almost always see log probabilities used in algorithms and machine learning papers, even though we are conceptually interested in the raw probabilities themselves.

By sampling a full episode, we get a pretty good idea of the true value of an action because we can see its downstream effects rather than just its immediate effect (which may be misleading due to randomness in the environment); this full episode sampling is under the umbrella of Monte Carlo approaches. But not all environments are episodic, and sometimes we want to be able to make updates in an incremental or *online* fashion, i.e., make updates at regular intervals irrespective of what is going on in the environment. Our deep Q-network did well in the non-episodic setting and it could be considered an online-learning algorithm, but it required an experience replay buffer in order to effectively learn.

The replay buffer was necessary because true online learning where parameter updates are made after each action is unstable due to the inherent variance in the environment. An action taken once may by chance result in a big negative reward, but in expectation (the average long-term rewards) it may be a good action—updating after a single action may result in erroneous parameter updates that will ultimately prevent adequate learning.

In this chapter, we will introduce a new kind of policy gradient method called *distributed advantage actor-critic* (DA2C) that will have the online-learning advantages of DQN without a replay buffer. It will also have the advantages of policy methods where we can directly sample actions from the probability distribution over actions, thereby eliminating the need for choosing a policy (such as the epsilon-greedy policy) that we needed with DQN.

5.1. Combining the value and policy function